

Module 4 **Evaluation Method**

- Why evaluation is essential in NLP
- Benchmark datasets and **gold-standard annotations**
- Why annotation reliability matters
- **Inter-annotator agreement**
- Observed agreement, expected agreement, and **Cohen's Kappa**
- The **confusion matrix**: TP, FP, FN, TN
- Precision, recall, and **F1-score**
- When accuracy can be misleading
- **Micro, macro, and weighted** averages
- Entity-level evaluation for NER

What does "evaluation" mean?



evaluation

/ɪˌvæljuˈeɪʃ(ə)n/

noun

the making of a judgement about the amount, number, or value of something; assessment.
"the evaluation of each method"

In Computer Science, synonymous to: **testing**

Why Do We Evaluate NLP Systems?

- Evaluation tells us how well a system is performing.
- Users care about whether a system is useful for a task.
- Developers care about whether a system is improving over time.
- Evaluation supports:
 - comparison
 - debugging
 - model selection
 - scientific reporting

Benchmark Data and Gold Standards

- Performance evaluation is often based on benchmark datasets.
- These datasets contain:
 - inputs
 - gold-standard labels
 - a fixed evaluation protocol
- Gold standards are usually created by human annotators following annotation guidelines.
- Reliable evaluation depends on reliable annotation.

Performance Evaluation

Often based on a **benchmark data set**

Organised around a community challenge/shared task

- Specific task is **defined**
- **Gold standard** data provided: training, development, test

Automated means for scoring submissions, whereby the following are compared:

response: system-generated annotations/predictions

reference: gold standard

The Kaggle logo, featuring the word "kaggle" in a blue, lowercase, sans-serif font.

Gold Standard Data

Time-consuming and costly to produce

Requires annotation instructions (**annotation guidelines**)

Annotations done by experts

- May need some training in linguistics
- **Multiple annotators** need to label the same samples (a subset)

Need to ensure that annotations are **reliable**

- Calculate **inter-annotator agreement**
- There might be disagreements

Inter-Annotator Agreement

- Human annotations are not always identical.
- Different annotators may interpret a guideline differently.
- Inter-annotator agreement measures how consistently humans label the same data.
- High agreement suggests the task and the guidelines are clearer and more reliable.

Observed Agreement: $P(a)$

- Observed agreement is the proportion of cases on which annotators agree.
- Example:
 - two annotators label 10 items
 - they agree on 7 items
- Then:
 - $P(a) = 7/10 = 0.70$
- Observed agreement does **not** correct for chance agreement.

Expected Agreement: $P(e)$

- Expected agreement estimates how much agreement would happen by chance.
- Suppose:
 - Annotator A assigns **Positive** to 6/10 items and **Negative** to 4/10
 - Annotator B assigns **Positive** to 5/10 items and **Negative** to 5/10
- Then:
 - chance agreement on Positive = $0.6 \times 0.5 = 0.30$
 - chance agreement on Negative = $0.4 \times 0.5 = 0.20$
- So:
 - $P(e) = 0.30 + 0.20 = 0.50$

Cohen's Kappa

(Exam)

- Cohen's Kappa corrects observed agreement for chance agreement.
- Formula:
 - $K = \frac{P(a) - P(e)}{1 - P(e)}$
- Using the previous example:
 - $P(a) = 0.70$
 - $P(e) = 0.50$
 - $K = (0.70 - 0.50) / (1 - 0.50) = 0.40$
- Kappa is more informative than raw agreement alone.

Kappa coefficient: Example

		Annotator 1		
		yes	no	total
Annotator 2	yes	31	1	32
	no	2	6	8
	total	33	7	40

$$P(a) = P(A1=yes, A2=yes) + P(A1=no, A2=no)$$

$$= (31/40) + (6/40)$$

$$= 0.925$$

Kappa coefficient: Example

		Annotator 1		
		yes	no	total
Annotator 2	yes	31	1	32
	no	2	6	8
	total	33	7	40

$$P(e) = P(A1=yes)*P(A2=yes) + P(A1=no)*P(A2=no)$$

$$= ((33/40) * (32/40)) + ((7/40) * (8/40))$$

$$= 0.695$$

Kappa coefficient: Example

		Annotator 1		
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$$= ((33/40) * (32/40)) + ((7/40) * (8/40))$$

$$= 0.695$$

Kappa coefficient: Interpretation

Landis and Koch, 1977

slight < 0.2 < fair < 0.4 < moderate < 0.6 < substantial < 0.8 < perfect

Grove et al., 1981 (psychiatric community)

0.6 < acceptable

Krippendorff, 1980

0.67 < tentative conclusions < 0.8 < definite conclusions

Rietveld and van Hout, 1993

0.4 < moderate < 0.6 < substantial < 0.8

Green, 1997

low < 0.4 < fair/good < 0.75 < high

Scoring

Precision: fraction of annotated items that are correct

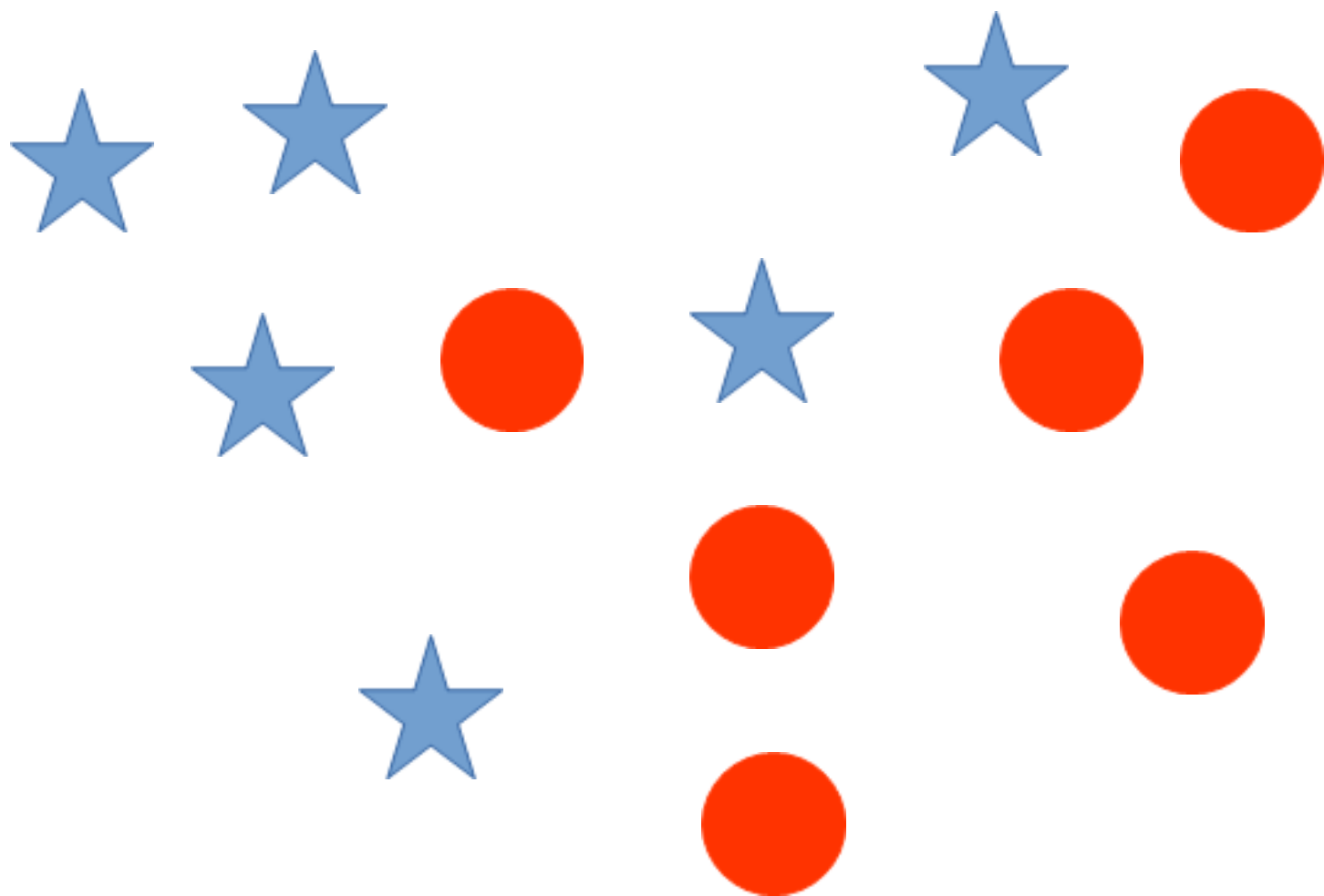
Recall: fraction of correct items that are annotated

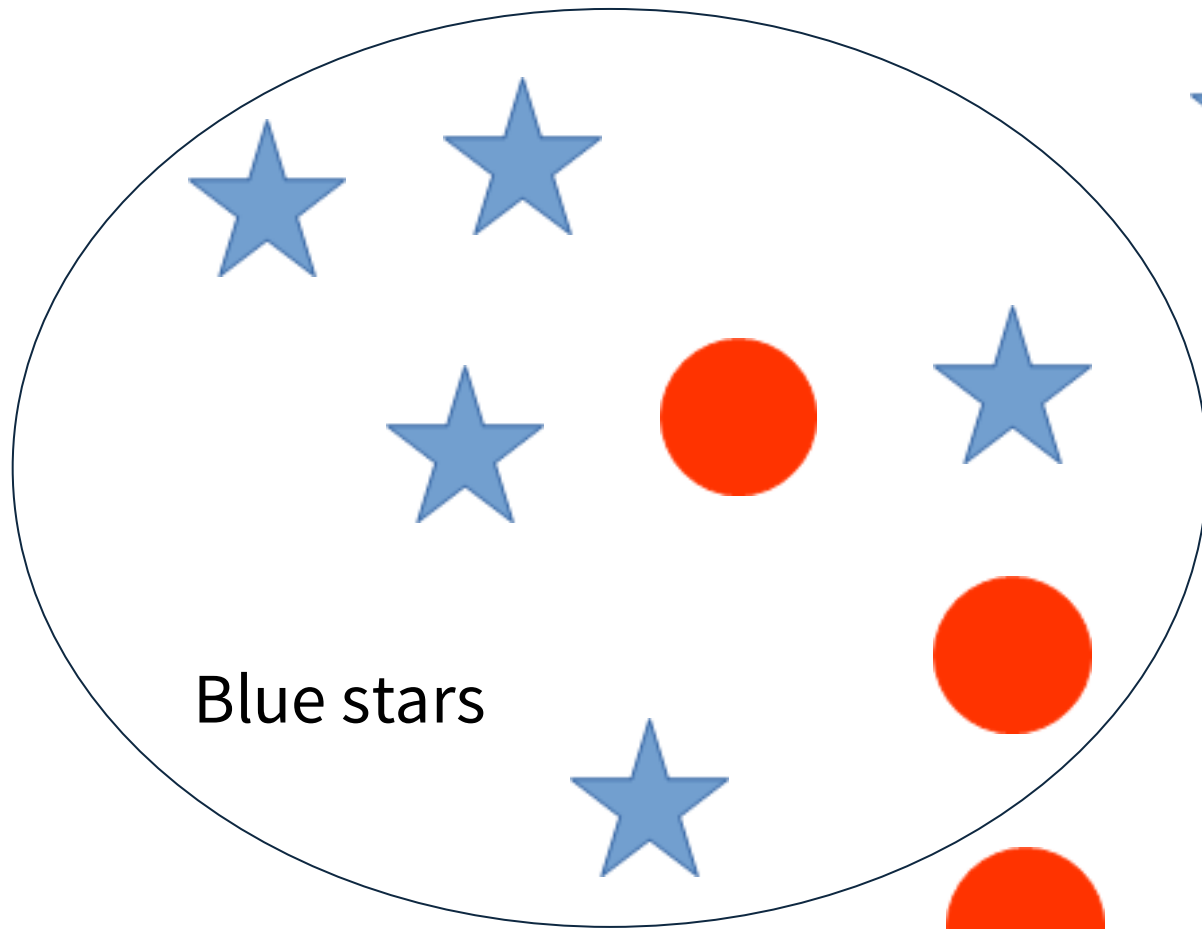
Confusion matrix:

	Correct	Not correct
Annotated	True positive (TP)	False positive (FP)
Not annotated	False negative (FN)	True negative (TN)

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

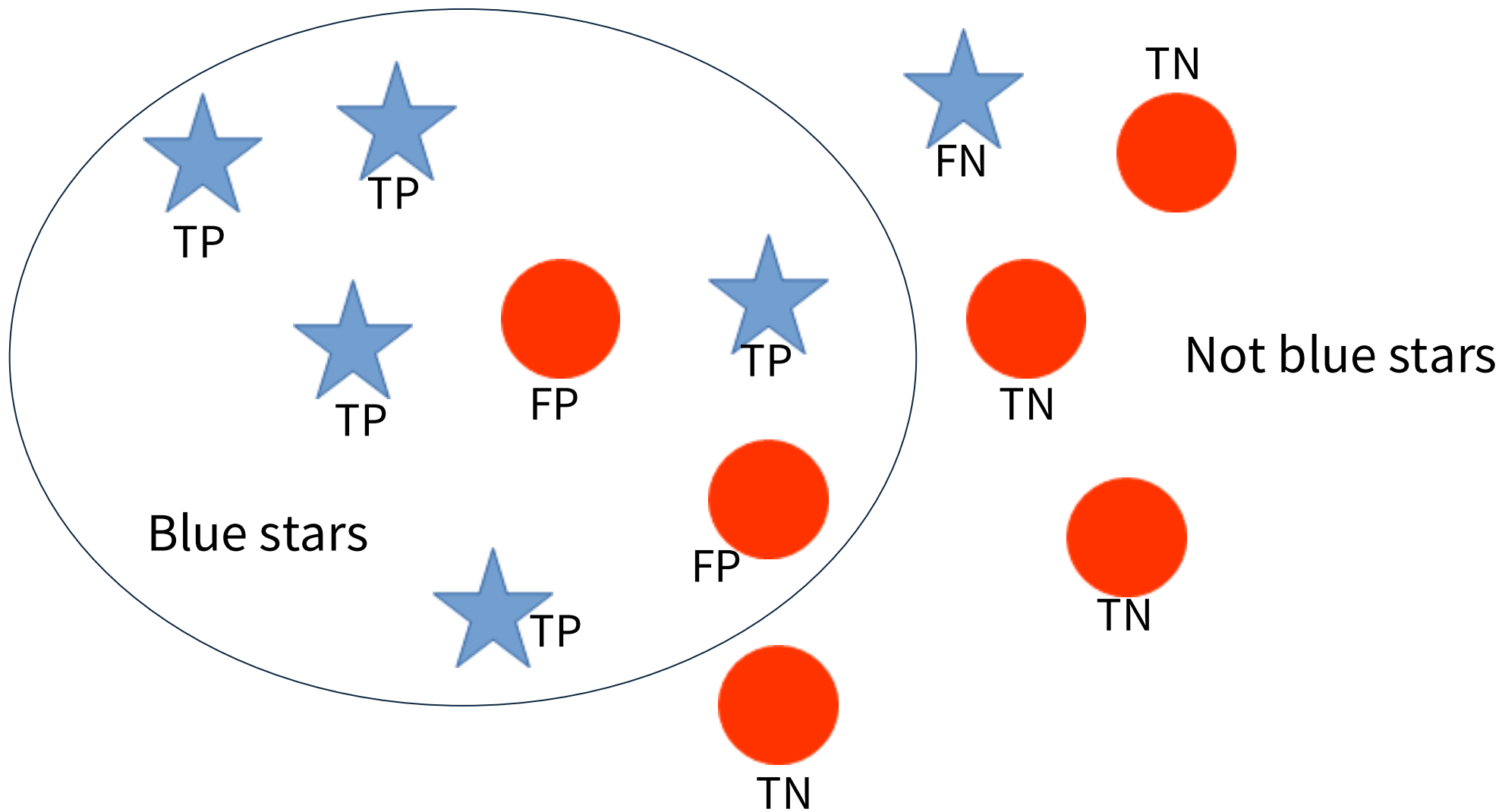
$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

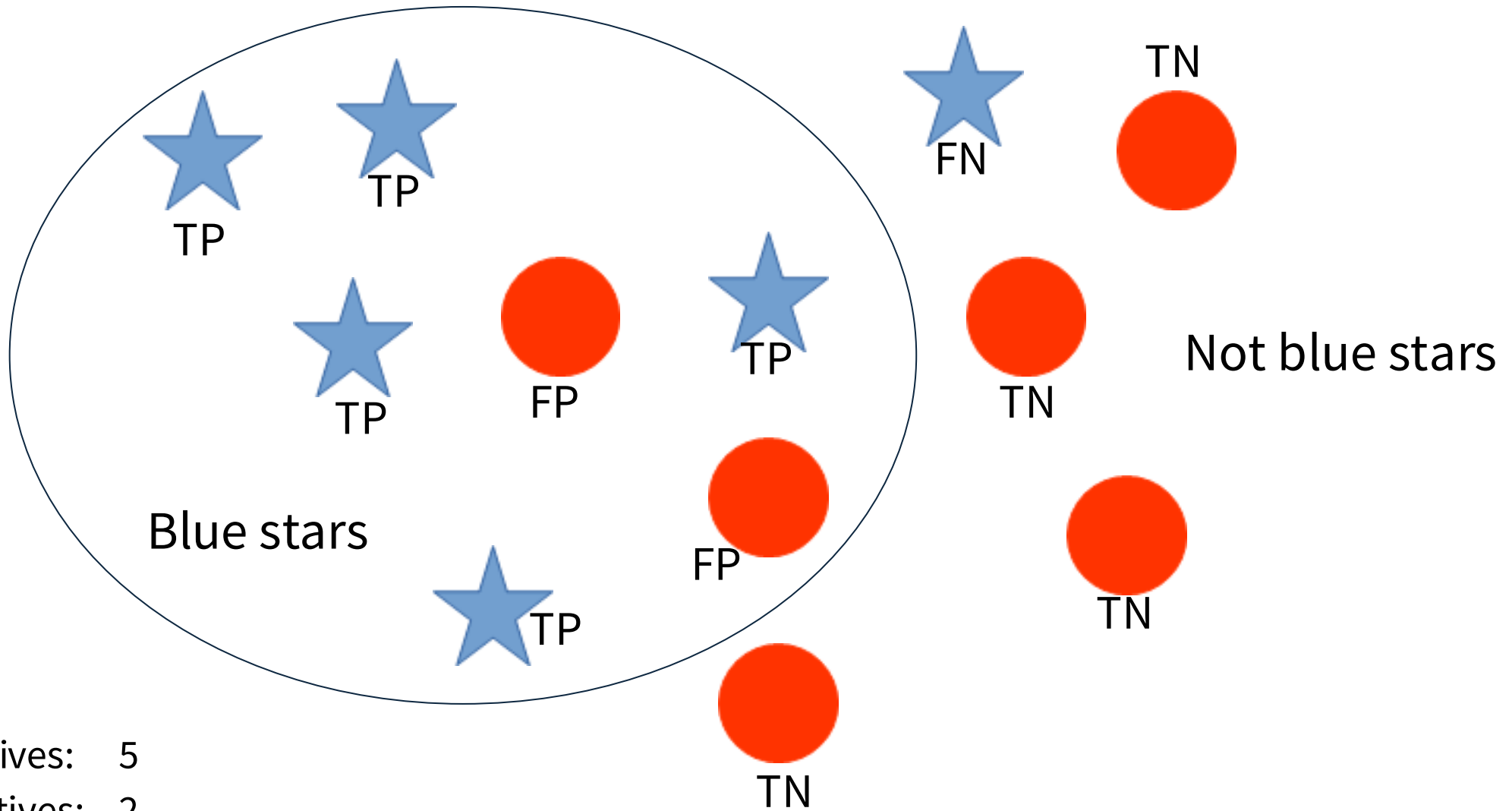




Blue stars

Not blue stars



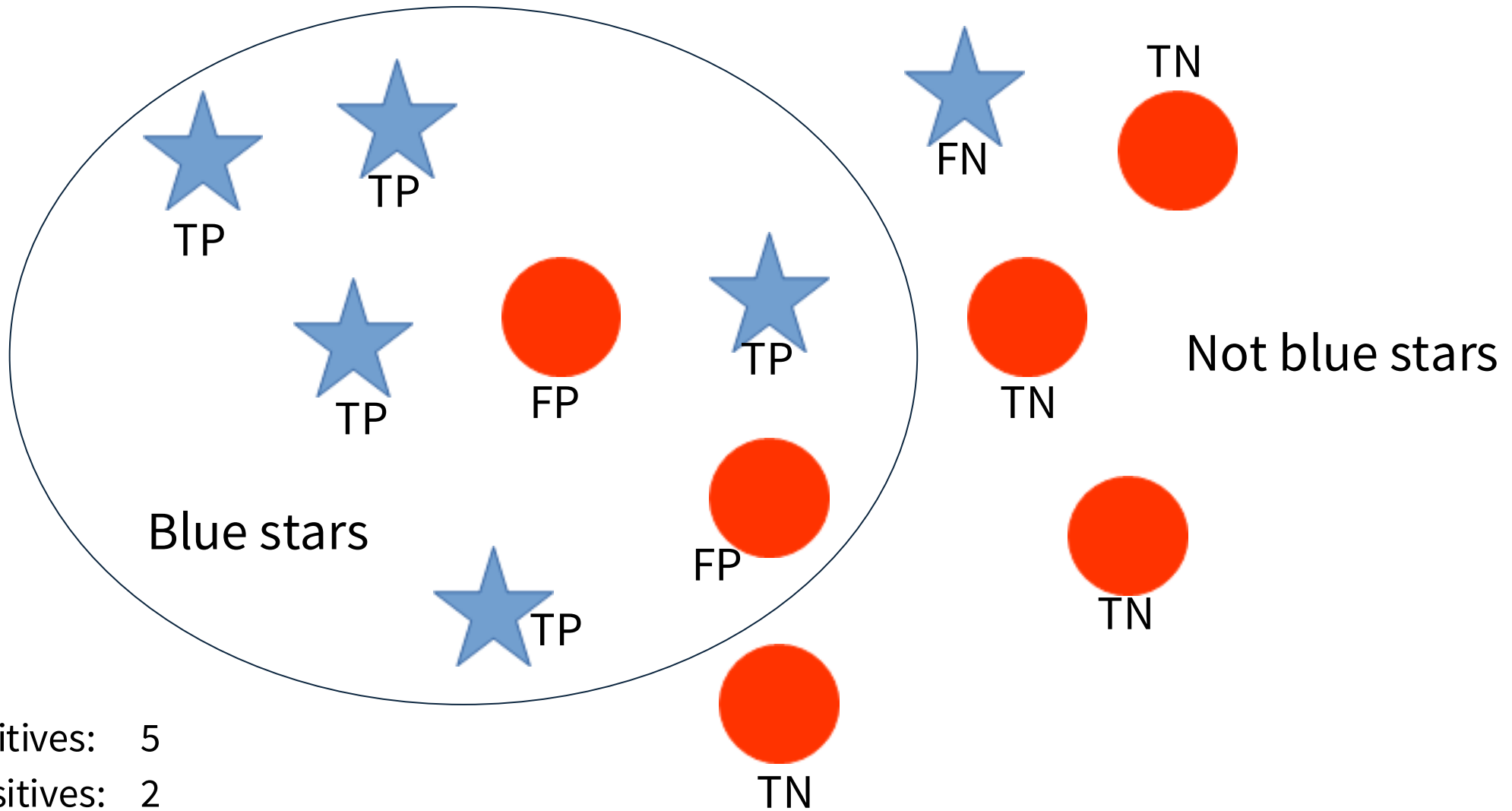


True Positives: 5

False Positives: 2

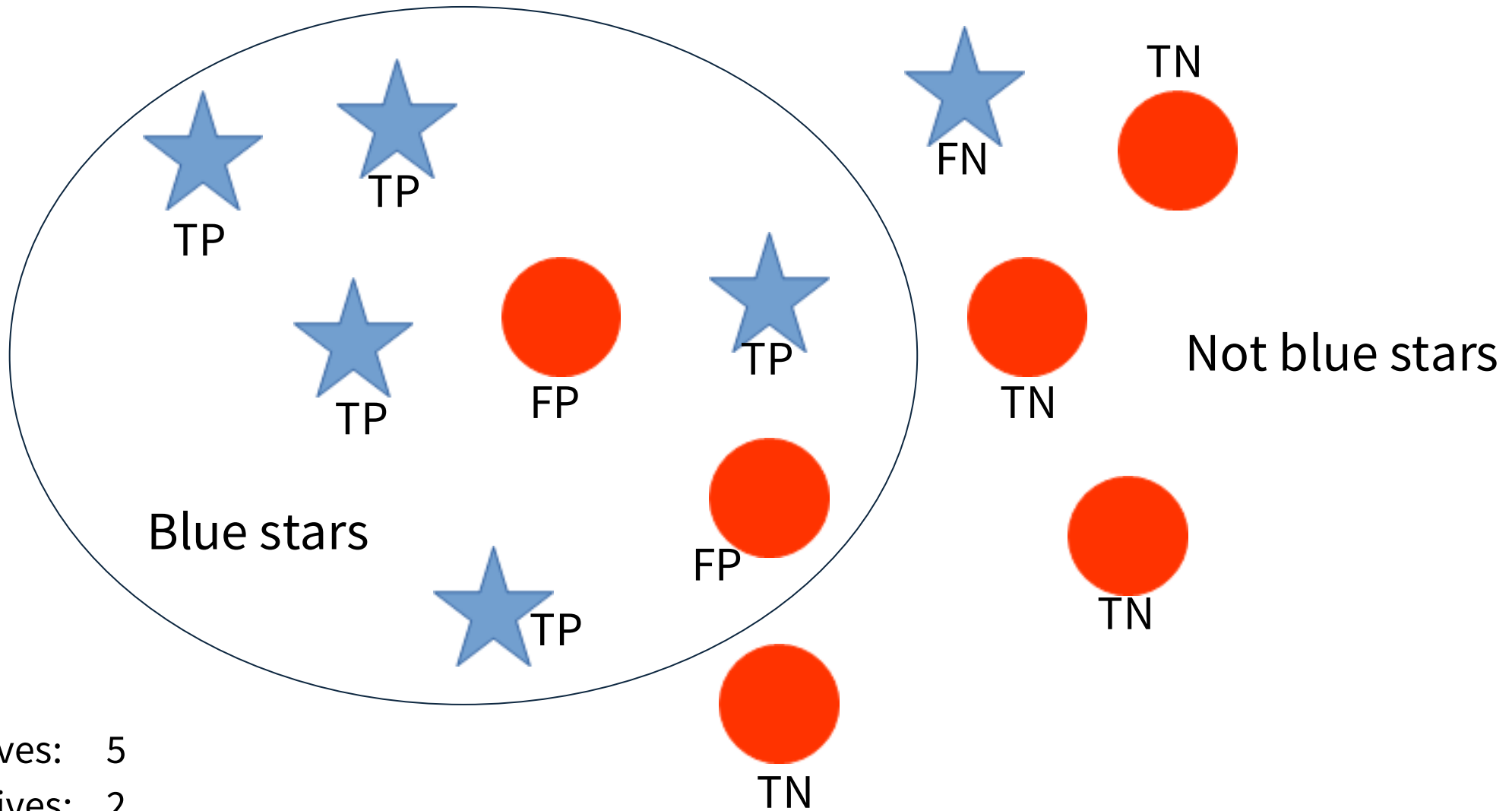
False Negatives: 1

True Negatives: 4



True Positives: 5
False Positives: 2
False Negatives: 1
True Negatives: 4

$$\text{Precision} = TP / (TP + FP) = 5 / (5 + 2) = 5 / 7 = 0.714$$



True Positives: 5

False Positives: 2

False Negatives: 1

True Negatives: 4

$$\text{Precision} = TP / (TP + FP) = 5 / (5 + 2) = 5 / 7 = 0.714$$

$$\text{Recall} = TP / (TP + FN) = 5 / (5 + 1) = 5 / 6 = 0.833$$

F-score (a.k.a. F-measure, F1-score)

Weighted harmonic mean

$$F_{\beta} = (\beta^2 + 1)PR / \beta^2 P + R$$

Usually balanced F1 measure is used, where $\beta=1$

$$F1 = 2PR / (P + R)$$

Harmonic mean is a more conservative average (truer picture)

F-score vs Arithmetic Mean

$$P = 0.714$$

$$R = 0.833$$

$$\text{Arithmetic Mean: } (0.714 + 0.833) / 2 = \mathbf{0.774}$$

$$\text{F-score: } 2 * 0.714 * 0.833 / (0.714 + 0.833) = \mathbf{0.769}$$

F-score vs Arithmetic Mean

$$P = 1$$

$$R = 0.15$$

$$\text{Arithmetic Mean: } (1 + 0.15) / 2 = \mathbf{0.575}$$

$$\text{F-score: } 2 * 1 * 0.15 / (1 + 0.15) = \mathbf{0.261}$$

What about Accuracy?

Measure of all correctly identified cases

$$Acc = (TP + TN) / (TP + TN + FP + FN)$$

Suitable if all classes are equally important. What if not?

Example: Hate vs Neutral

Use F-score

Multiple Categories: Micro vs Macro-averaging

Category	TPs	FPs	FNs	Precision	Recall
Person	78	5	33	0.94	0.70
Location	20	3	2	0.87	0.91

How do we report combined performance for Person and Location?

Option 1: **Macro-averaging** -- Simply get the average

$$P_{\text{Person+Location}} = (0.94+0.87)/2 = 0.91$$

$$R_{\text{Person+Location}} = (0.70+0.91)/2 = 0.81$$

$$F1_{\text{Person+Location}} = (2*0.91*0.81)/(0.91+0.81) = 0.86$$

Multiple Categories: Micro vs Macro-averaging ↓ (Exam)

Category	TPs	FPS	FNs	Precision	Recall
Person	78	5	33	0.94	0.70
Location	20	3	2	0.87	0.91

How do we report combined performance for Person and Location?

Option 2: **Micro-averaging** -- Pool together the TPs, FPS and FNs

$$P_{\text{Person+Location}} = (78+20)/((78+20)+(5+3)) = 0.92$$

$$R_{\text{Person+Location}} = (78+20)/((78+20)+(33+2)) = 0.74$$

$$F1_{\text{Person+Location}} = (2*0.92*0.74)/(0.92+0.74) = 0.82$$

Which is better?

Category	TPs	FPs	FNs	Precision	Recall
Person	78	5	33	0.94	0.70
Location	20	3	2	0.87	0.91

Macro-averaging does not consider class imbalance; micro-averaging is less sensitive to imbalance

Weighted average:

average weighted by the number of true instances for each class

Confusion Matrix for evaluating automated predictions

- For automated systems, evaluation often starts with a confusion matrix.
- For a positive class:
 - **TP**: predicted positive, actually positive
 - **FP**: predicted positive, actually negative
 - **FN**: predicted negative, actually positive
 - **TN**: predicted negative, actually negative
- Many common metrics are derived from these four counts.

	Positive class (Response)	Negative class (Response)
Positive class (Reference)	True Positives (TP)	False Negatives (FN)
Negative class (Reference)	False Positives (FP)	True Negatives (TN)

Worked NER Example: Counting Entities

(Exam)

- Tokens:
 - *Dr | Maya | Chen | joined | Acme | Labs | in | Bristol | .*
- Gold tags:
 - *B-PER I-PER I-PER O B-ORG I-ORG O B-LOC O*
- Predicted tags:
 - *B-PER I-PER O O B-ORG I-ORG O O O*
- Entity counts:
 - **PER**: TP = 0, FP = 1, FN = 1
 - **ORG**: TP = 1, FP = 0, FN = 0
 - **LOC**: TP = 0, FP = 0, FN = 1
- The person prediction is wrong because the predicted span is incomplete.

Gold entities

- Maya Chen = PER
- Acme Labs = ORG
- Bristol = LOC

Predicted entities

- Maya Chen = PER
- Acme = ORG
- Bristol = LOC

Counts

- PER: TP = 1, FP = 0, FN = 0
- ORG: TP = 0, FP = 1, FN = 1
- LOC: TP = 1, FP = 0, FN = 0

Key point:

A partially correct span does **not** count as a true positive.

Precision and Recall

- **Precision** asks:
 - Of the items predicted as positive, how many are correct?
 - $Precision = TP / (TP + FP)$
- **Recall** asks:
 - Of the truly positive items, how many did the system find?
 - $Recall = TP / (TP + FN)$
- Precision and recall capture different aspects of performance.

The F1 Score

- F1 combines precision and recall using the harmonic mean.
- Formula:
 - $F1 = \frac{2PR}{P+R}$
- F1 is high only when both precision and recall are high.
- It is often preferred when the positive class is important and class balance is uneven.

Worked Example: Computing F1

- Suppose a spam detector produces:
 - TP = 18
 - FP = 6
 - FN = 9
- Then:
 - Precision = $18 / (18 + 6) = 0.75$
 - Recall = $18 / (18 + 9) = 0.67$
 - F1 = $2 \times 0.75 \times 0.67 / (0.75 + 0.67) \approx 0.71$
- This shows the full calculation from counts to F1.

When Accuracy Can Mislead

- Accuracy measures the proportion of all correct decisions.
- Formula:
 - $Accuracy = (TP + TN) / (TP + TN + FP + FN)$
- Accuracy can be misleading on imbalanced datasets.
- Example:
 - if 95% of items are negative, always predicting negative gives 95% accuracy
 - but recall for the positive class is 0
- In such cases, F1 is often more informative.

Micro, Macro, and Weighted Averages

- For multi-class problems, we often report combined performance.
- **Macro average:**
 - compute the metric for each class
 - then average across classes equally
- **Micro average:**
 - pool all TP, FP, and FN first
 - then compute the metric once
- **Weighted average:**
 - average class-level metrics using class frequency as weights
- The choice affects how strongly rare classes influence the final score.

Worked Example: Micro-Averaged F1 for NER

Using the previous example:

- PER: TP = 1, FP = 0, FN = 0
- ORG: TP = 0, FP = 1, FN = 1
- LOC: TP = 1, FP = 0, FN = 0

Now pool all counts across entity types:

- **Total TP = 2**
- **Total FP = 1**
- **Total FN = 1**

- $Precision_{micro} = \frac{2}{2+1} = 0.667$ ✓

- $Recall_{micro} = \frac{2}{2+1} = 0.667$ ✓

- $F1_{micro} = \frac{2 \times 0.667 \times 0.667}{0.667 + 0.667} = 0.667$ ✓

- **Interpretation:**

Micro-averaging gives one overall score after combining all entity decisions across categories.

Module 5. **Open Information Extraction**

- What Open Information Extraction is
- How OpenIE differs from predefined relation extraction
- A brief view of classical and neural OpenIE approaches

What Is Open Information Extraction?

- Open Information Extraction (OpenIE) extracts relational facts from text without relying on a fixed predefined schema.
- Input:
 - raw sentences or documents
- Output:
 - relation tuples such as *(argument, relation, argument)*
- OpenIE aims to turn unstructured text into machine-readable knowledge.

What an OpenIE Extraction Looks Like

- Sentence:
 - *Albert Einstein published the theory of relativity in 1915.*
- Example extraction:
 - *(Albert Einstein; published; the theory of relativity)*
 - *(Albert Einstein; published in; 1915)*
- OpenIE relies heavily on earlier NLP steps:
 - POS tagging
 - parsing
 - entity recognition

A Short View of OpenIE Methods and Limits

- Classical methods used:
 - patterns
 - dependency structure
 - clause splitting
- Later work introduced supervised and neural approaches.
- Common challenges include:
 - ambiguity
 - synonymy
 - coreference
 - trustworthiness of extracted facts
- In this module, OpenIE is a compact application case rather than the main assessment focus.